

Informedness of the gradient-informed single-site proposal

Why it scales as $\Delta t/\sqrt{T}$, how Δt is chosen,
and what the $\Delta t = 460$ vs. $\Delta t = 63$ runs showed

`my_extendedSampling` / `softmax`

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Abstract

The sampler in `classes/rate_sampler.py` proposes one substitution per Monte Carlo move by reading off which amino acid a gradient-informed Gaussian displacement “points at”. How reliably that displacement follows the gradient’s own preference—its *informedness*—is controlled by Δt , but *not* by Δt alone: the governing combination is $\Delta t/\sqrt{MT}$. This note derives that scaling from the proposal construction, records the measured informedness curves for both reference sequences, explains why Δt is chosen at a deliberately *partial* informedness of $p_{\text{match}} \approx 0.15$ rather than as large as possible, and reports what the paired $\Delta t = 460$ / $\Delta t = 63$ Boltzmann-validation runs on `zero_polymer` ($L = 566$) established. The short version of the last point: both runs reproduce the exactly-enumerated Boltzmann distribution, so the Δt error that motivated the pair cost test *coverage*, not correctness.

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1 Scope and sources

Everything below is either derived from the code as it stands or measured in a run whose output file is committed. Source of truth for each:

- Proposal construction: `classes/rate_sampler.py`, method `_step`.
- Pointing probability: `utils/proposals.py`, function `log_pointing_prob`.
- Informedness curves: `tests/test_informedness_vs_dt.py`; outputs `protein_g/informedness_vs_dt_results.txt` and `zero_polymer/informedness_vs_dt_results.txt`.
- Boltzmann validation: `tests/validate_boltzmann_toy.py`; outputs `zero_polymer/validate_boltzmann_results_dt460.txt` and `..._dt63.txt`.
- Chronology and the mistake that prompted this note: `softmax/DEVLOG.txt`, entries dated 2026-10-02 and 2026-10-03.

Quantities that are *estimated* rather than measured are flagged as such at the point of use; there are two (the effective sample size in §6.4, and the p_{match} interpolations in Table 2).

2 The proposal, and what “informedness” means

2.1 One move

A sequence of length L is held as per-site logits over the $K = 25$ -letter vocabulary `SEQUENCE_USED_VOCAB`, mean-centred along the class axis to gauge-fix the translation invariance of the softmax. One move:

1. Pick one site s uniformly from $\{0, \dots, L-1\}$. Uniform and state-independent, so it cancels exactly from the Metropolis–Hastings ratio and needs no correction term.
2. At that site only, draw a displacement $\delta x \in \mathbb{R}^K$ (§2.2) and propose the substitution it points at, *excluding* the site’s current residue and the five non-canonical codes `X`, `B`, `U`, `Z`, `O`. That leaves

$$K_c = 25 - 5 - 1 = 19$$

competitors. (K_c is 20 in the one edge case where the site already sits on a non-canonical code, since then there is no current residue to remove from the canonical set.)

3. Accept or reject with an exact Metropolis–Hastings correction using the true recomputed energy of the candidate and the true reverse proposal probability, so the target is $\exp(-U/T)$ exactly rather than to first order.

Note that step 2 always proposes a genuine change: “staying” is decided by rejection in step 3, never by the proposal.

2.2 The displacement

The displacement is one HMC leapfrog half-step with fresh momentum, which for a single step is the same object as a MALA/Langevin step:

$$\delta x = \frac{\Delta t p}{M} - \frac{\Delta t^2}{2M} \nabla U, \quad p \sim \mathcal{N}(0, MT I), \quad (1)$$

with $\nabla U = \partial U / \partial(\text{logits at } s)$ obtained by relaxing *only* site s through $\text{softmax}(\cdot / T_{\text{sftm}})$ while every other site is held at its exact one-hot value. Equation (1) makes δx Gaussian, and this is the pair of quantities everything downstream depends on:

$$\delta x \sim \mathcal{N}(\mu, \sigma^2 I), \quad \boxed{\mu = -\frac{\Delta t^2}{2M} \nabla U, \quad \sigma = \Delta t \sqrt{\frac{T}{M}}} \quad (2)$$

So the drift grows as Δt^2 while the noise grows only as Δt . That asymmetry is the entire mechanism by which Δt buys informedness, and §4 is just this observation carried through carefully.

2.3 Informedness, defined

Write j^* for the competitor the gradient alone would pick,

$$j^* = \arg \max_{j \in \text{competitors}} \mu_j = \arg \min_{j \in \text{competitors}} (\nabla U)_j, \quad (3)$$

the second equality because $\mu = -\Delta t^2 / (2M) \cdot \nabla U$ and $\Delta t^2 / (2M)$ is a *positive scalar*. This gives the first of two facts that the diagnostic `test_informedness_vs_dt.py` was built to separate:

- (A) **The gradient’s choice is independent of Δt .** Scaling μ by a positive scalar cannot change its $\arg \max$. So *which* residue the gradient prefers, and therefore how good that preference is, does not depend on Δt at all. Δt cannot improve the gradient’s taste.
- (B) **How often the realised draw obeys that choice does depend on Δt .** The actual proposal is the $\arg \max$ of δx , which is μ plus noise. Define

$$p_{\text{match}} = \mathbb{P} \left(\arg \max_j \delta x_j = j^* \right), \quad (4)$$

the probability that one realised move follows the gradient’s own pick. This is what Δt controls, and it is what “informedness” means throughout this note.

Conflating (A) and (B) was the original error that this decomposition fixed: an earlier diagnostic swept Δt looking for improvements in proposal *quality*, which by (A) cannot exist. The two limits of (B) are

$$p_{\text{match}} \longrightarrow \frac{1}{K_c} = \frac{1}{19} \approx 0.0526 \quad (\Delta t \rightarrow 0), \quad p_{\text{match}} \longrightarrow 1 \quad (\Delta t \rightarrow \infty), \quad (5)$$

i.e. a uniform pick among the 19 competitors at small Δt , and a deterministic, greedy gradient-descent pick at large Δt . Note the small- Δt limit is *uniform*, not “stay put”: nothing in (1) references the current residue’s identity, which is why an earlier design that tried to make small Δt mean “few sites change” could not work and was abandoned.

3 The pointing probability $a_{i \rightarrow j}$

3.1 Closed form

p_{match} , and more generally the proposal probability needed for the Metropolis–Hastings ratio, is the probability that a particular component of a Gaussian vector is the largest. For target j among competitors k , with a common σ (true here, since Δt , T , M are global scalars):

$$a_{i \rightarrow j} = \mathbb{P}(\delta x_j > \delta x_k \ \forall k \neq i, j) = \int_{-\infty}^{\infty} \phi(u) \prod_{k \neq i, j} \Phi(\alpha_{kj} + u) \, du, \quad \alpha_{kj} = \frac{\mu_j - \mu_k}{\sigma}, \quad (6)$$

with ϕ , Φ the standard normal density and CDF. With constant σ this is a Thurstone/probit discrete-choice model—the probit analogue of the softmax-based proposal used by Gibbs–With–Gradients. It is an exact probability, not merely something proportional to one: summing (6) over all admissible j at a fixed site gives 1, which is checked numerically in `tests/test_proposals.py`.

3.2 How it is evaluated

`log_pointing_prob` evaluates (6) by Gauss–Hermite quadrature entirely in log space, using $n_{\text{quad}} = 40$ nodes:

$$\log a_{i \rightarrow j} = \log \sum_q \exp \left[\log w_q + \sum_{k \neq j} \log \Phi(\alpha_{kj} + u_q) \right], \quad (7)$$

computed with `torch.special.log_ndtr` and `logsumexp` so that no intermediate underflows even when α is large. The current residue is removed by *slicing* it out of the tensor being maximised over, not by adding a large negative penalty—a deliberate change from an earlier version, since a finite penalty could in principle be approached by Δt^2 -scaled gradients at large Δt , whereas a sliced-out class is structurally absent.

3.3 The one approximation

Equation (6) treats the K components of δx as independent. They are not exactly: mean-centring the momenta and logits induces the covariance $\sigma^2(I - J/K)$ rather than $\sigma^2 I$, a pairwise correlation of $-1/(K-1) \approx -0.05$ for $K = 20$. This is a deliberate, small, documented approximation—the exact calculation would be a $(K-1)$ -dimensional orthant integral with no closed form. It is worth knowing it is there, because it is the one place the proposal probability is not exact; it is not, however, a threat to the sampler’s target distribution, since the *same* formula is used for both the forward and reverse direction and the Metropolis–Hastings correction only needs their ratio to be the ratio of the probabilities actually used to propose.

4 Why informedness scales as $\Delta t / \sqrt{T}$

4.1 The derivation

This is the central result of the note, and it is three lines. Inspecting (6), the *only* way Δt , T and M enter $a_{i \rightarrow j}$ is through the standardised gaps $\alpha_{kj} = (\mu_j - \mu_k)/\sigma$. Substituting (2):

$$\begin{aligned} \alpha_{kj} &= \frac{\mu_j - \mu_k}{\sigma} = \frac{-\frac{\Delta t^2}{2M} [(\nabla U)_j - (\nabla U)_k]}{\Delta t \sqrt{T/M}} \\ &= \frac{\Delta t}{2\sqrt{MT}} [(\nabla U)_k - (\nabla U)_j] = \frac{\Delta t g_{kj}}{2\sqrt{MT}}, \quad g_{kj} \equiv (\nabla U)_k - (\nabla U)_j. \end{aligned} \quad (8)$$

The Δt^2 of the drift divides by the Δt of the noise, leaving one power of Δt ; the temperature enters only through σ , as $\sqrt{T}$. Hence:

$$p_{\text{match}} \text{ depends on } (\Delta t, T, M, \nabla U) \text{ only through } \frac{\Delta t}{\sqrt{MT}} \text{ and the gradient gaps } g_{kj}. \quad (9)$$

4.2 Three consequences

1. An exact invariance. Any two settings with the same $\Delta t/\sqrt{MT}$ have *identical* informedness. With $M = 1$ throughout this project, holding p_{match} fixed while changing temperature requires

$$\Delta t' = \Delta t \sqrt{\frac{T'}{T}}. \quad (10)$$

This is an exact statement about the proposal, not an approximation or a fitted relation.

2. Hotter sampling is less informed at fixed Δt . Raising T inflates σ without touching μ , so the same Δt buys strictly less informedness at higher temperature. There is a certain physical justice to this—a hotter chain should explore more freely—but it means Δt is *not* a transferable constant, and any quoted Δt is meaningless without the T it was measured at.

3. The trap, which we fell into. `test_informedness_vs_dt.py` hard-codes its own $T = 2.0$ as a module constant, so every p_{match} it reports is a $T = 2$ number. Reading $\Delta t = 63$ off that table and using it for a run at $T = 107$ —which is what the first version of the queued command did—changes α by $1/\sqrt{107/2} = 1/7.31$ and lands the proposal essentially back on the uniform floor. In the equivalence of (10),

$$\Delta t = 63 \text{ at } T = 107 \quad \equiv \quad \Delta t = \frac{63}{\sqrt{107/2}} = 8.6 \text{ at } T = 2, \quad (11)$$

and at $\Delta t \approx 8.6$ the measured $T = 2$ curve gives $p_{\text{match}} \approx 0.06$ against a uniform baseline of 0.0526. The correct value for the intended working point is

$$\Delta t = 63 \times \sqrt{107/2} = 63 \times 7.3144 \approx 460. \quad (12)$$

This is now recorded in a comment on `DTS_TO_SWEEP` so that the next reader does not repeat it. For the record, `protein_g`'s long-standing $\Delta t = 1300$ needed no rescaling only because its runs used $T = 2.0$, the same temperature the informedness script assumes—the coupling was invisible for as long as only one temperature was ever used.

5 Choosing Δt

5.1 Why not simply make Δt large?

By (B), large Δt drives $p_{\text{match}} \rightarrow 1$, i.e. a deterministic greedy pick. That is undesirable for three independent reasons.

- 1. The gradient's pick is not that good.** By (A), Δt cannot improve it. Measured top-1 accuracy—how often j^* is the true energetic optimum among the 19 competitors, by brute force—is 2/10 on `protein_g` and 10/30 on `zero_polymer` measured away from the energy floor. So a fully greedy proposal would commit to a substitution that is wrong roughly two thirds of the time, with no mechanism left for exploring the alternatives.

2. **A deterministic proposal cripples the Metropolis–Hastings correction.** As $p_{\text{match}} \rightarrow 1$ the proposal becomes a near-deterministic map, the reverse probability $a_{j \rightarrow i}$ for the move that undoes it becomes very small, and the Hastings ratio $a_{j \rightarrow i}/a_{i \rightarrow j}$ swings wildly. The chain’s mobility collapses.
3. **Specific sites saturate at wildly different rates.** The measured spread across sites is large—at $\Delta t = 1000$ on `zero_polymer`, p_{match} ranges from 0.40 to 1.00 depending on the site. A Δt chosen for the mean already pins some sites at 1.0.

Equally, Δt too small gives a uniform proposal: correct, but it throws away the gradient information the whole construction exists to exploit, and the Hastings term becomes ≈ 0 because $a_{i \rightarrow j} \approx a_{j \rightarrow i} \approx 1/K_c$.

5.2 The working point

The target is therefore deliberately *partial* informedness—meaningfully above uniform, nowhere near greedy. The convention this project settled on is

$$p_{\text{match}} \approx 0.15, \tag{13}$$

roughly $3\times$ the uniform baseline of $1/19 = 0.0526$ and far from saturation. It leaves the gradient a real but non-binding say, keeps all 19 competitors reachable within a realistic move budget, and keeps both $a_{i \rightarrow j}$ and $a_{j \rightarrow i}$ comfortably away from their extremes so the Hastings correction stays well-conditioned.

5.3 Measured curves

Table 1 is the measured $p_{\text{match}}(\Delta t)$ for both references, both at $T = 2.0$, pooled over 10 brute-forced sites. Note these are *closed-form* evaluations of (6) at the real measured gradients, not Monte Carlo estimates: once ∇U is known at a site (one forward/backward pass), sweeping arbitrarily many Δt values costs nothing further. This is why extending the sweep downward was free.

The two curves are displaced by about a factor of 20 in Δt : at a given Δt , `zero_polymer` is far more informed. By (8) with T and M equal, that displacement is entirely attributable to larger gradient gaps g_{kj} on the longer sequence, which is consistent with the independently measured $\approx 18\times$ larger gradient components there.

5.4 Recommended practice

1. Never quote Δt without the T it belongs to.
2. Changing T requires rescaling Δt by $\sqrt{T'/T}$, (10), or informedness silently moves.
3. Re-measure Table 1 for a new reference sequence. The curve depends on the gradient gaps, which are a property of the sequence and the energy, and they moved by $20\times$ between these two references.
4. Check the per-site (min, max) spread, not just the mean. A mean of 0.15 can hide sites already saturated at 1.0.

6 What $\Delta t = 460$ vs. $\Delta t = 63$ showed

6.1 The experiment

`validate_boltzmann_toy.py` makes the sampler’s target distribution exactly checkable. It frees exactly two sites (30 and 52), pins all 564 others at the reference, and enumerates every one of the

Δt	protein_g ($L = 56$)		zero_polymer ($L = 566$)	
	mean p_{match}	(min, max)	mean p_{match}	(min, max)
10	—	—	0.0647	(0.056, 0.090)
20	—	—	0.0788	(0.058, 0.137)
50	—	—	0.1285	(0.068, 0.313)
100	—	—	0.2132	(0.084, 0.562)
200	—	—	0.3479	(0.121, 0.818)
300	0.0725	(0.060, 0.099)	0.4436	(0.160, 0.933)
600	0.0954	(0.068, 0.160)	0.6229	(0.275, 0.999)
1000	0.1280	(0.080, 0.256)	0.7447	(0.403, 1.000)
1300	0.1524	(0.089, 0.331)	0.7930	(0.470, 1.000)
2000	0.2047	(0.111, 0.483)	0.8507	(0.503, 1.000)
4000	0.3197	(0.178, 0.743)	0.9156	(0.520, 1.000)
8000	0.4669	(0.300, 0.948)	0.9506	(0.541, 1.000)
16000	0.6266	(0.391, 1.000)	0.9581	(0.581, 1.000)
32000	0.7691	(0.472, 1.000)	0.9659	(0.659, 1.000)

Table 1: Measured $p_{\text{match}}(\Delta t)$, both at $T = 2.0$, $M = 1$, pooled over 10 sites. Uniform baseline $1/19 = 0.0526$. The **protein_g** sweep was never run below $\Delta t = 300$; the **zero_polymer** range was extended down to $\Delta t = 10$ precisely because the old range turned out to sit entirely above its working point.

target p_{match}	Δt at $T = 2$	Δt at $T = 107$	regime
≈ 0.0526	—	—	uniform floor, no gradient information
0.10	33	240	weakly informed
0.15	63	460	the working point
0.25	127	931	strongly informed
0.50	394	2885	approaching greedy
0.79	1281	9372	effectively greedy

Table 2: Working points for **zero_polymer**, with the $T = 107$ column obtained from the $T = 2$ column by the exact rescaling (10), factor $\sqrt{107/2} = 7.3144$. The $T = 2$ values are linear interpolations within Table 1 and so carry interpolation error; the rescaling itself is exact.

$20^2 = 400$ states by brute force, giving the exact Boltzmann distribution $p_{\text{exact}} \propto \exp(-U/T)$. It then runs the real `_step` logic restricted to those two sites and compares the empirical visit histogram against p_{exact} .

Both runs used $T = 107$ (chosen from the participation-ratio calibration: the inherited $T = 2.0$ puts $p = 1.0000$ on a single state and the chain cannot move), $M = 1$, 20 000 moves, 4000 discarded as burn-in. They differ *only* in Δt , and by §4 that means they differ only in informedness:

run	Δt	informedness at $T = 107$
informed	460	$p_{\text{match}} \approx 0.15$, the intended working point
uniform	63	$p_{\text{match}} \approx 0.06$, essentially the 1/19 floor

The $\Delta t = 63$ run exists because it was the erroneous value in the first version of the queued command. Running it alongside the corrected one turned the mistake into a controlled comparison: the same sampler, same energy, same temperature, with the gradient information effectively switched off.

6.2 Results

	$\Delta t = 460$ (informed)	$\Delta t = 63$ ($\approx$ uniform)
acceptance rate	0.3060	0.2893
$\text{KL}(p_{\text{emp}} \parallel p_{\text{exact}})$	0.0316	0.0338
total variation distance	0.0787	0.0848
distinct states visited	224/400	223/400
mean U under p_{exact}	+185.20	+185.20
mean U under p_{emp}	+185.63	+187.63
deviation	+0.44 (0.24%)	+2.43 (1.31%)

Table 3: Boltzmann validation on `zero_polymer`, $T = 107$, 20 000 moves, 16 000 post-burn-in samples over 400 enumerated states. $\log 400 = 5.99$ is the rough scale against which to read the KL.

6.3 Reading the result

Both pass; this validates the sampler at $L = 566$. The top states come back in essentially the right order and the right proportions, and mean U —the most robust single summary, being an average over the whole distribution rather than a per-state frequency—agrees to 0.24% at the informed Δt . Neither run shows the decisive failure signature the script warns about: mass piled on states the exact distribution calls unlikely, or the true top states missed. This was the last outstanding safety check and the only one that exercises accept/reject end to end.

The Δt error cost coverage, not correctness. This is the substantive methodological lesson of the pair. A near-uniform proposal still targets the same Boltzmann distribution—the Metropolis–Hastings correction does not care how good the proposal is, only that its forward and reverse probabilities are computed consistently. So the $\Delta t = 63$ run converged perfectly respectably. But with $a_{i \rightarrow j} \approx a_{j \rightarrow i} \approx 1/K_c$, the Hastings term

$$\log a_{j \rightarrow i} - \log a_{i \rightarrow j} \approx 0, \quad (14)$$

which means that run never meaningfully exercised the reverse-probability pass—the second backward pass at the candidate point, and the asymmetry correction it feeds. That is the most

intricate part of the sampler and the part most likely to harbour a subtle bug, and it is the reason this test exists. Had only the $\Delta t = 63$ run been done, it would have been reported as a pass while leaving that machinery untested. The informed run is the one that actually tests it.

Informedness helps, but modestly, and not via acceptance. The informed run is better on every metric, but only slightly, and acceptance barely moved (0.306 vs. 0.289). Informedness is not acting as an acceptance-rate lever here. The natural reading is that on a 400-state problem with a ≈ 41 -state effective support, a uniform proposal is already a decent explorer, so there is little headroom; the informed proposal’s advantage should be expected to matter much more on the full 566-site problem, where the competitor set per move is the same 19 but the state space is vastly larger. That expectation is untested.

6.4 Is the residual discrepancy bias or noise?

The KL values are small but not zero, and it is worth being precise about whether that residue is bias. The participation ratio of p_{exact} at $T = 107$ is

$$k_{\text{eff}} = \left(\sum_i p_i^2 \right)^{-1} = 41.4 \quad (15)$$

effective states out of 400. For n *independent* samples from the exact distribution, the expected KL of the empirical histogram is approximately

$$\mathbb{E}[\text{KL}] \approx \frac{k_{\text{eff}} - 1}{2n} = \frac{40.4}{2 \times 16000} = 0.00126, \quad (16)$$

about $25\times$ smaller than observed. But MCMC samples are not independent. Inverting the same expression for the effective sample size actually implied:

$$\text{ESS} \approx \frac{k_{\text{eff}} - 1}{2 \text{KL}} = \begin{cases} 639 & (\Delta t = 460) \\ 598 & (\Delta t = 63) \end{cases} \implies \tau_{\text{auto}} = \frac{16000}{\text{ESS}} \approx 25\text{--}27 \text{ moves.} \quad (17)$$

An autocorrelation time of ≈ 25 moves is entirely ordinary for a two-site chain at 30% acceptance: each site is re-proposed roughly every two moves and the proposal is accepted about 30% of the time, so the state decorrelates over tens of moves. Both KL values are therefore what a *correct* sampler limited by chain autocorrelation looks like. This is an estimate, not a measurement— τ_{auto} was inferred from the KL rather than computed from the chain.

The falsifiable check. If the sampler is unbiased, the KL is pure finite-sample noise and must fall as $1/n$:

$$\text{KL}(n) \propto \frac{1}{n} \implies \text{KL}(40000 \text{ moves}) \approx 0.016. \quad (18)$$

A KL that instead plateaus near 0.03 would indicate genuine bias. Until that run exists, the defensible claim is *no detectable bias at this resolution*, not *unbiased*. The check is cheap:

```
cd softmax/code
T=107 DT=460 MOVES=40000 python -u tests/validate_boltzmann_toy.py \
    2>&1 | tee zero_polymer/validate_boltzmann_dt460_M40000.txt
```

7 Caveats and open items

1. **The T window is from a two-site subproblem.** The usable range $T \approx 32\text{--}322$, and hence the $\Delta t \approx 460$ derived from it, were obtained with 564 of 566 residues pinned at the reference. A full-length chain explores a far wider energy range, so that window is a starting bracket, not a final answer. `zero_polymer` needs its own temperature sweep.

2. **Informedness curves are floor measurements.** The gradients behind Table 1 were computed at a seq_A only 5 mutations from the reference—0.9% drift at $L = 566$, where the energy sits essentially at its floor. Measured at a comparable drift to **protein_g** ($\approx 8.8\%$), the gradient’s behaviour changes substantially: top-1 rises from 2/10 to 10/30 and the mean true ΔU at the gradient’s pick goes from +474 to -360 . Gradient *gaps*, and hence the whole $p_{\text{match}}(\Delta t)$ curve, should be re-measured in the bulk before Table 2 is trusted for a production run.
3. **The independence approximation** of §2 is the one inexactness in $a_{i \rightarrow j}$. Unquantified against the exact orthant integral.
4. **Per-site saturation** is wide (Table 1), so a single global Δt is necessarily a compromise. Whether a per-site Δt would help, and whether it could be made to preserve detailed balance, is unexplored.
5. **1/n scaling of the KL** (§6.4) is the one outstanding validation item.